CS430/910: Foundations of Data Analytics

Clustering 3 | Dr Ayse Sunar

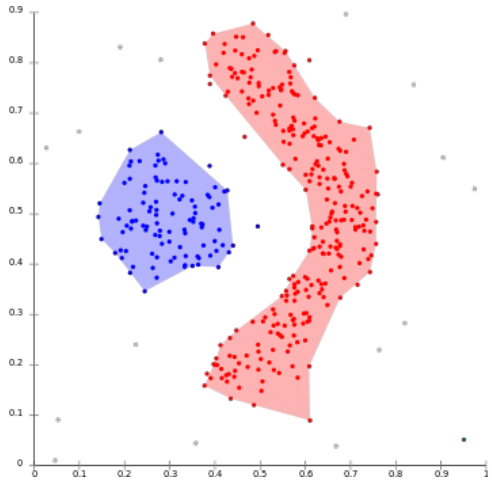
Objectives

- To understand the concept of clustering and clustering methods
- To see different clustering methods and objectives:
  - **k-means**
  - **k-medoids**
  - **Hierarchical Agglomerative Clustering**
  - **DBSCAN**
- To use tools (Weka) to cluster data and study the results

Part D: Clustering Algorithms 3 & Clustering Evaluation

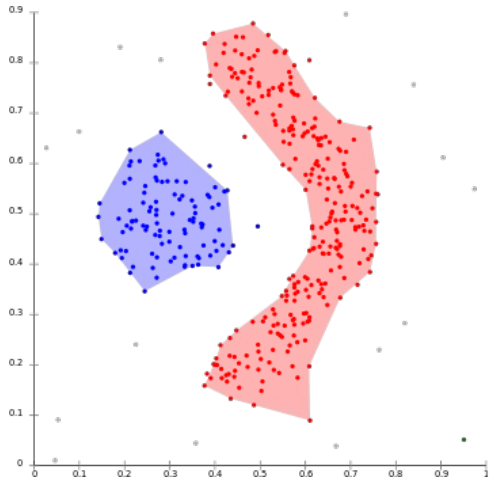
Density-Based Clustering Methods

- Many other clustering methods cluster based on the distance between objects.
  - Leads to spherical clusters, struggles with arbitrary shaped clusters.
- General idea: Continue growing a cluster as long as the nearby neighbourhood is greater than some threshold.
  - Good for filtering out outliers!



Density-Based Clustering Methods

- Density-based methods seek clusters that are locally dense
  - Dense meaning there are many data points per unit volume.
  - Separated by sparse regions of space.
  - Potentially allows finding complex cluster shapes.
  - May be more robust to noise
- Use a number of density parameters to determine when to extend a cluster.



DBSCAN

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is an example of a density-based clustering method.
- It finds *core objects* (objects with dense neighbourhoods).
- Uses two parameters:

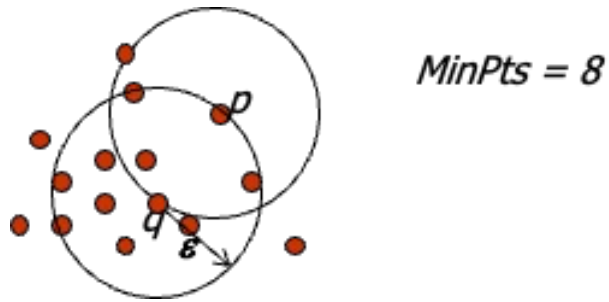
  1. ϵ (epsilon/eps): the radius to search for neighbours, based on a distance function $d()$.
  2. $minPts$: the minimum number of points in the neighbourhood of a point to extend the current cluster.

DBSCAN

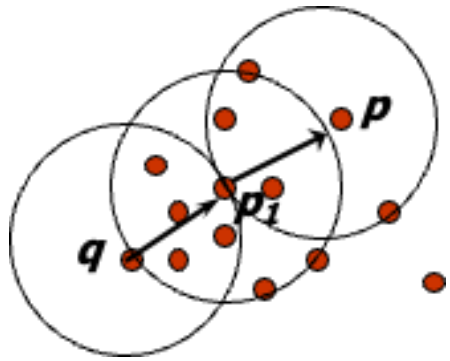
- The ϵ-neighbourhood (N_ϵ) of an object $\mathbf{o}$ is the space within a radius ϵ centred at object $\mathbf{o}$.
  - $N_\epsilon(\mathbf{q}) = \{\mathbf{p} \in data | d(\mathbf{p}, \mathbf{q}) \leq \epsilon\}$
- The density of a neighbourhood is the number of objects within the neighbourhood.
- An object is a *core object* if at least $minPts$ objects are within its ϵ-neighbourhood.

DBSCAN

- **Directly Density-Reachable**: For a **core** object $\mathbf{q}$ and an object $\mathbf{p}$, $\mathbf{p}$ is directly density-reachable from $\mathbf{q}$ (under ϵ, $minPts$) if $\mathbf{p}$ is within the ϵ-neighbourhood of $\mathbf{q}$.
  - Or, $\mathbf{p} \in N_\epsilon(\mathbf{q})$ and $|N_\epsilon(\mathbf{q})| \geq minPts$



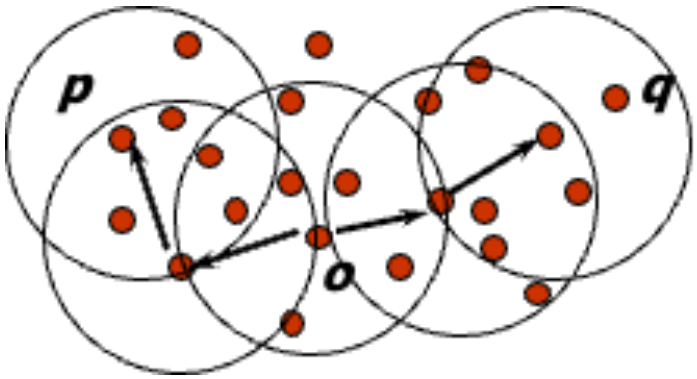
- **Density-Reachable**: An object $\mathbf{p}$ is density-reachable from an object $\mathbf{q}$ (under ϵ, $minPts$) if there exists a chain of objects $\mathbf{p_1}, \mathbf{p_2}, \ldots, \mathbf{p_n}$, such that $\mathbf{p_1} = \mathbf{q}$, $\mathbf{p_n} = \mathbf{p}$ and $\mathbf{p_{i+1}}$ is directly density-reachable from $\mathbf{p_i}$.



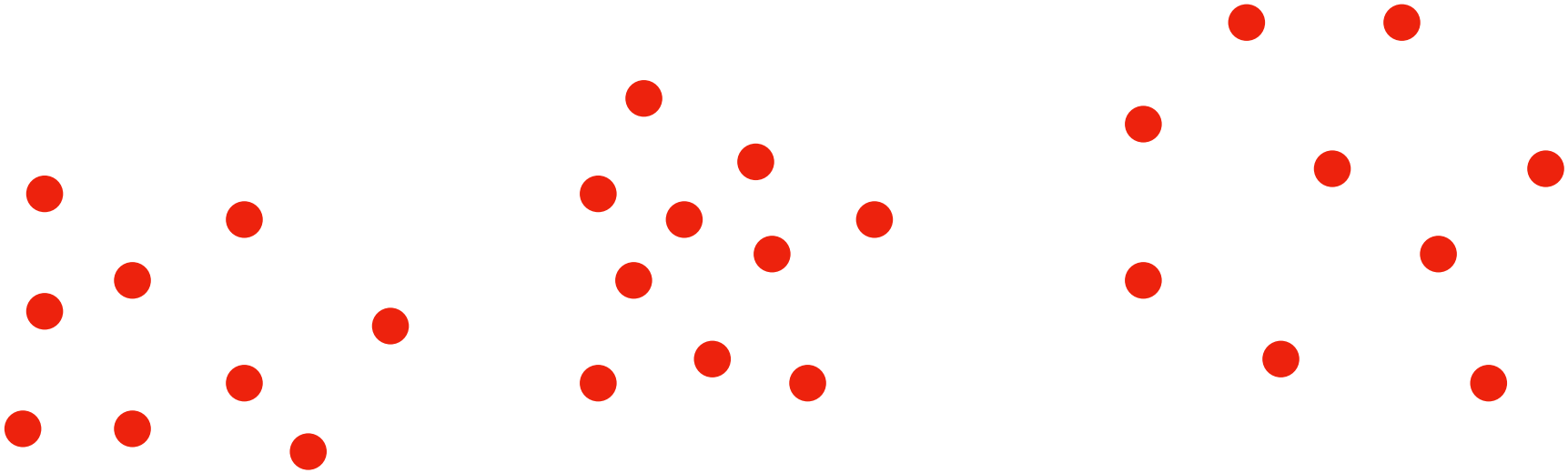
- Not symmetric! q might be a dense core point while **p** might not meet the density criteria so **p** can be density-reachable from **q**, the reverse isn't true.

DBSCAN

- **Density-Connected**: An object $\mathbf{p}$ is density-connected to an object $\mathbf{q}$ (under ϵ, $minPts$) if there is an object $\mathbf{o}$ such that both $\mathbf{p}$ and $\mathbf{q}$ are density-reachable from $\mathbf{o}$.
  - Symmetric!



DBSCAN

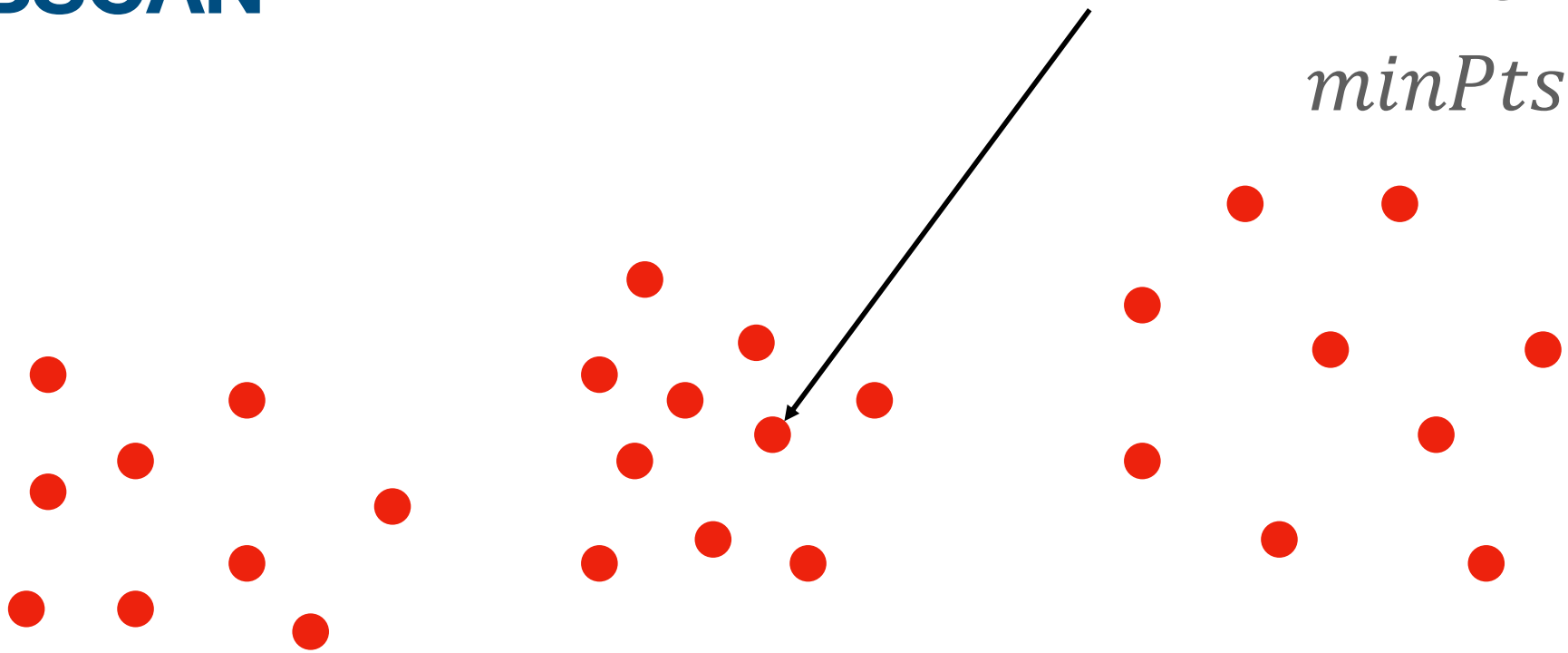


Start with this data.

DBSCAN

$\epsilon = 5$

$minPts = 3$

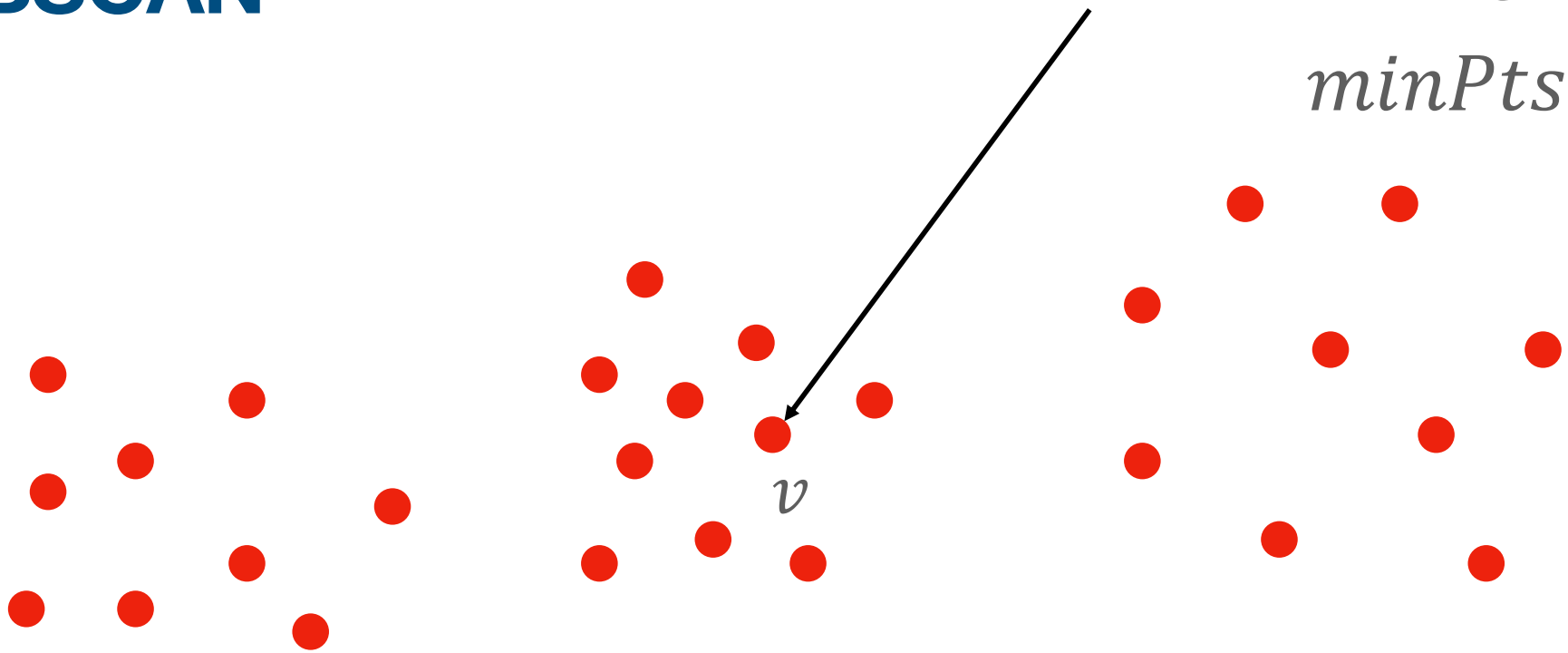


Select random object (lets call it $\mathbf{p}$).

DBSCAN

$\epsilon = 5$

$minPts = 3$

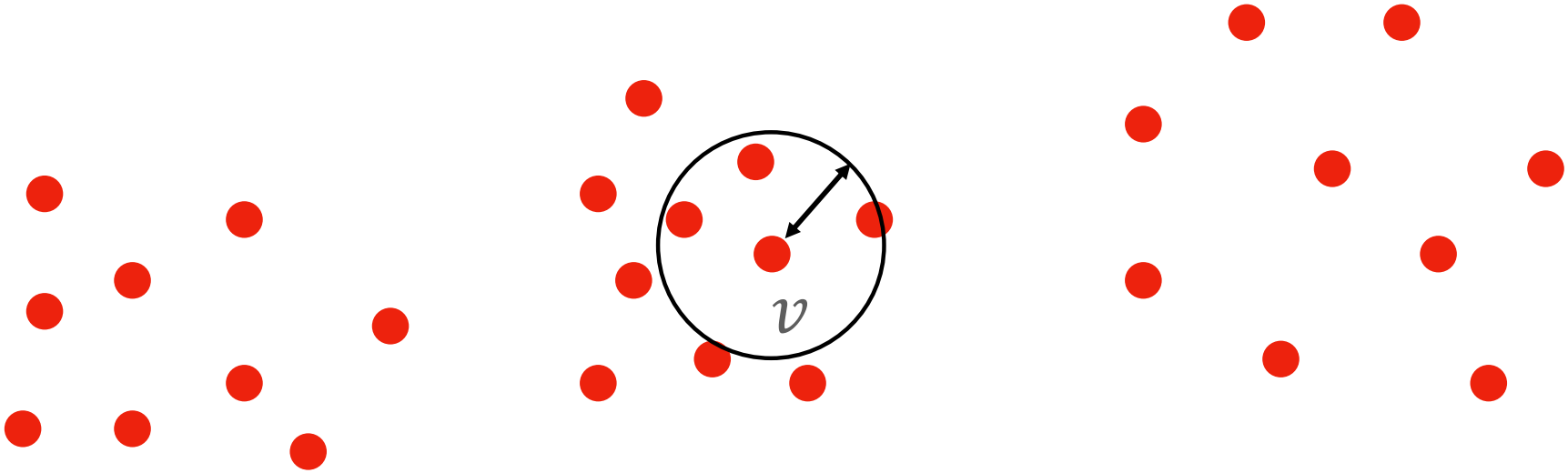


Mark $\mathbf{p}$ as visited.

DBSCAN

$\epsilon = 5$

$minPts = 3$



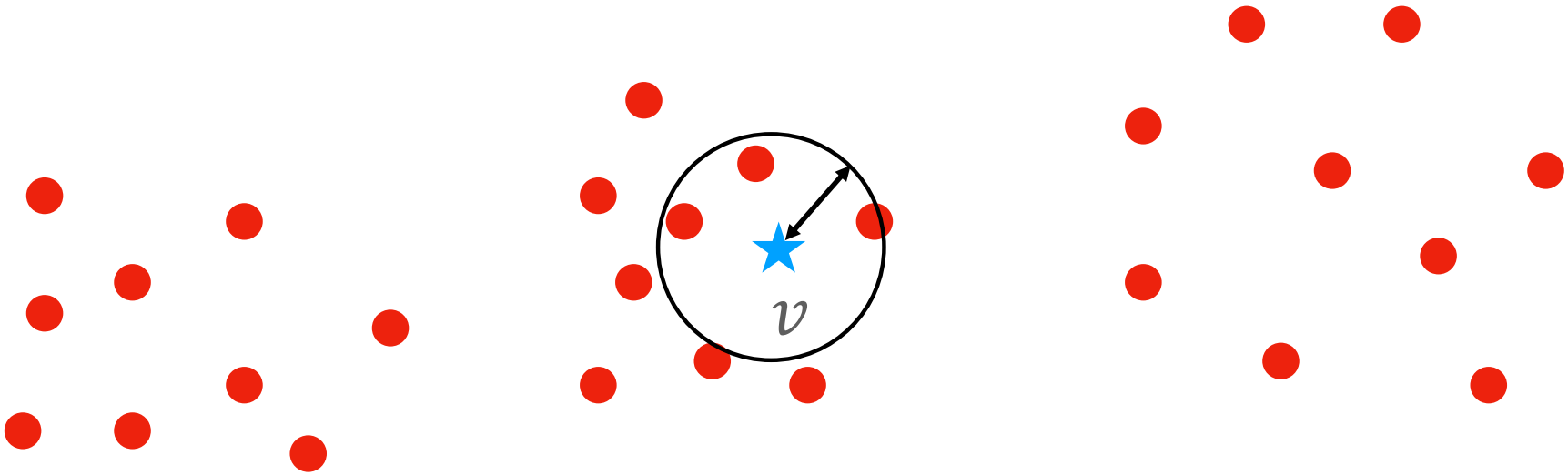
Are 3 objects (including original) within a distance of 5 of $\mathbf{p}$? Yes

(Otherwise we would mark $\mathbf{p}$ as noise and select a different unvisited random object)

DBSCAN

$\epsilon = 5$

$minPts = 3$

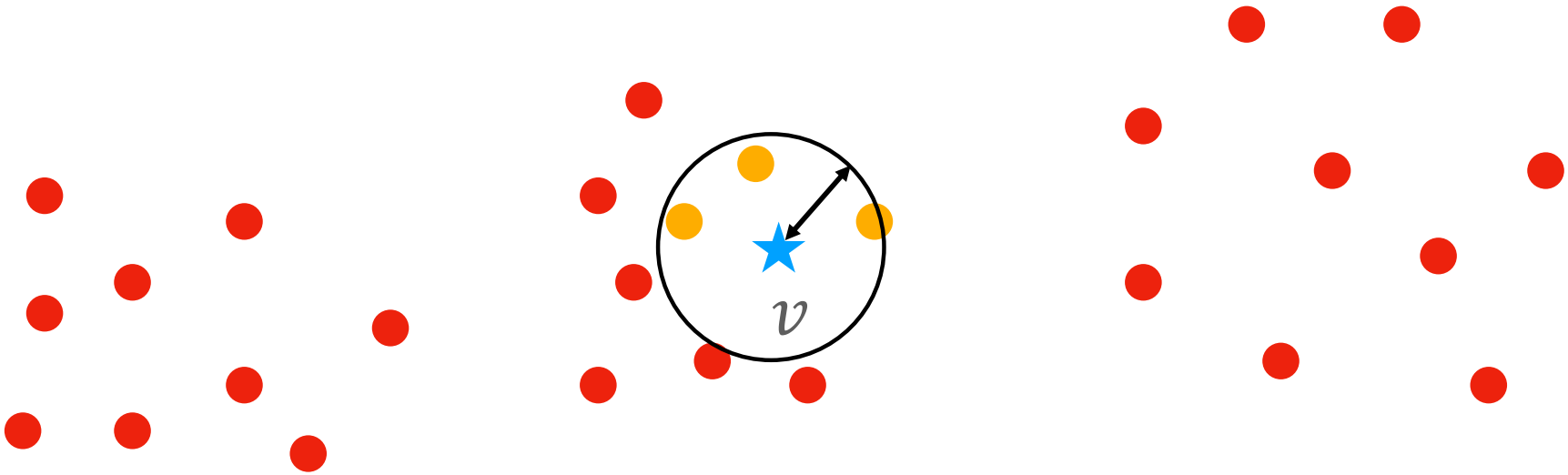


Create a new cluster C, and add $\mathbf{p}$ to it.

DBSCAN

$\epsilon = 5$

$minPts = 3$

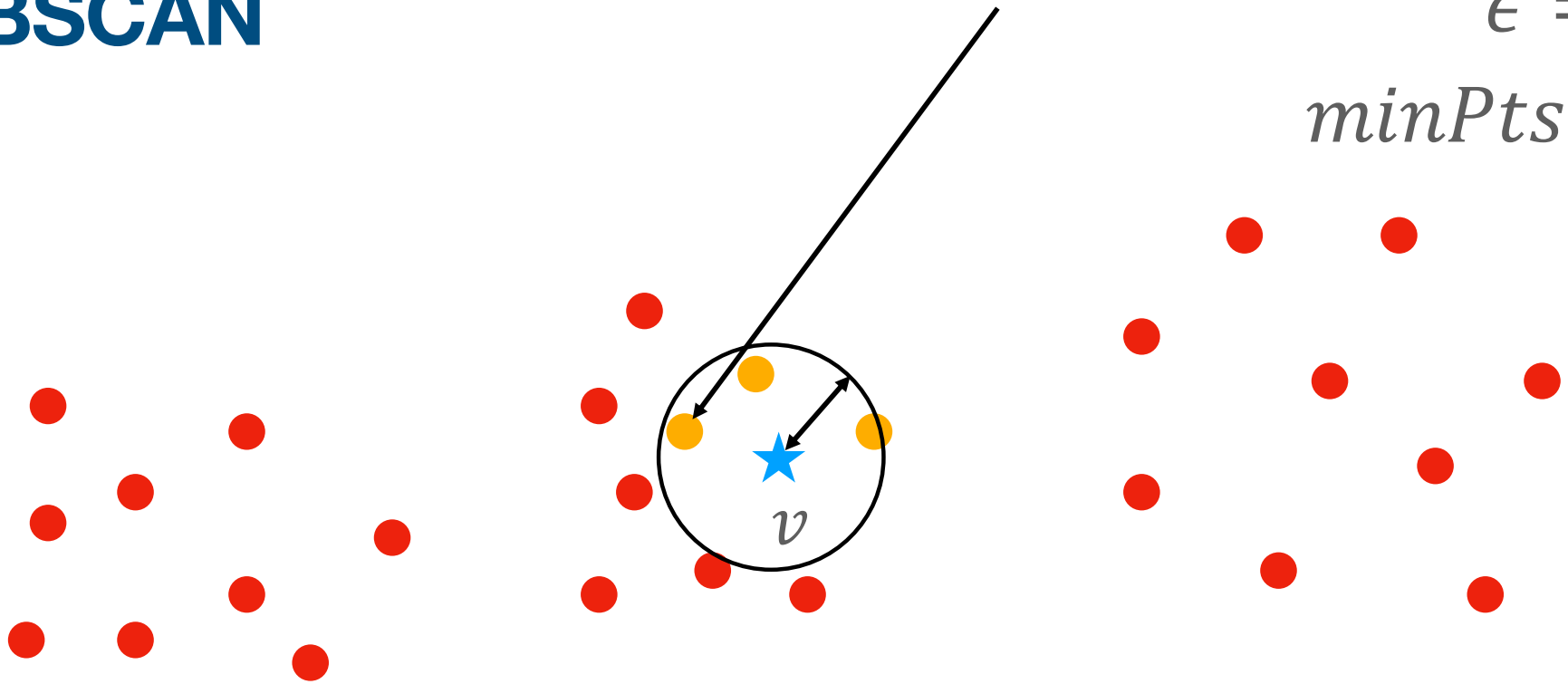


Let N be the set of objects in the ϵ-neighbourhood of $\mathbf{p}$.

DBSCAN

$\epsilon = 5$

$minPts = 3$

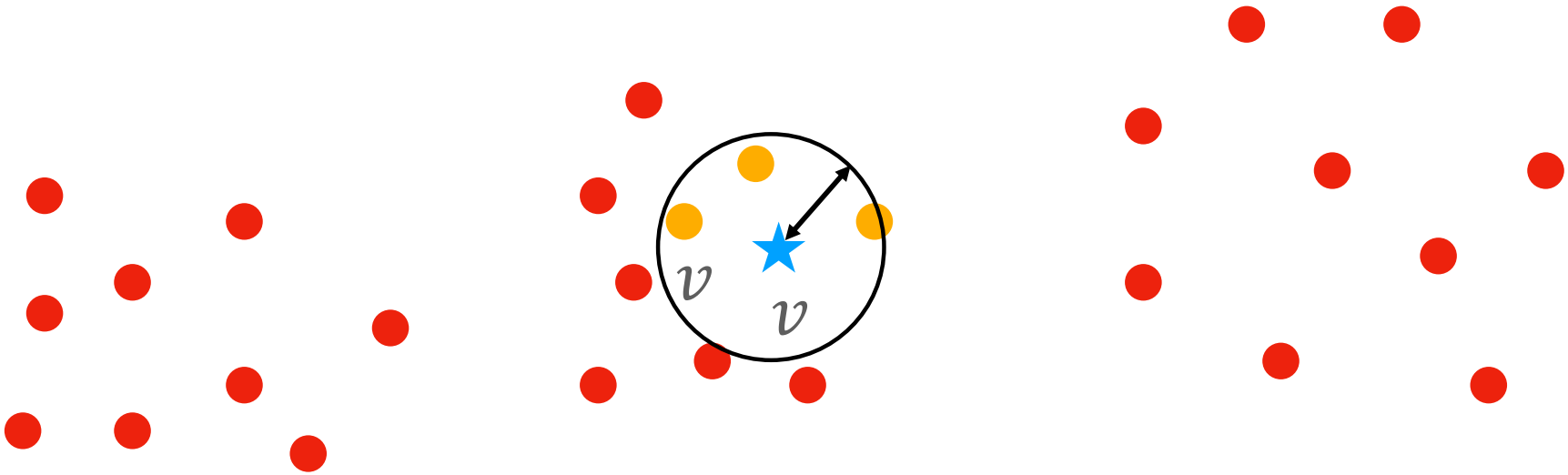


For each object $\mathbf{p}'$ in N…
(let's focus on the left one for now…)

DBSCAN

$\epsilon = 5$

$minPts = 3$

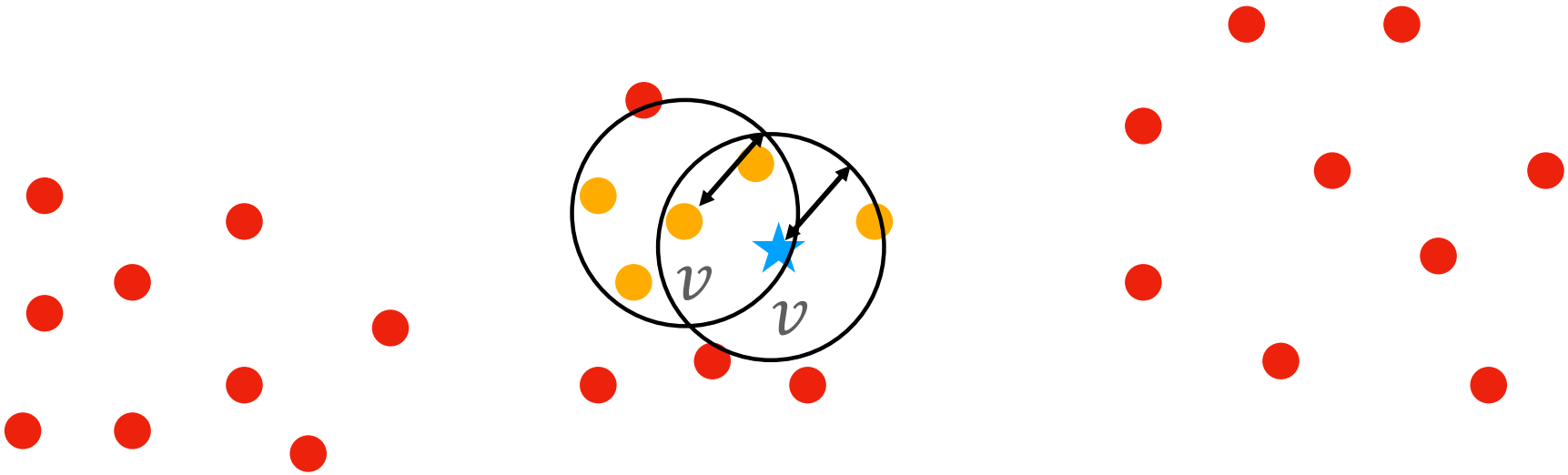


If $\mathbf{p}'$ is unvisited, 1) mark it as visited.

DBSCAN

$\epsilon = 5$

$minPts = 3$

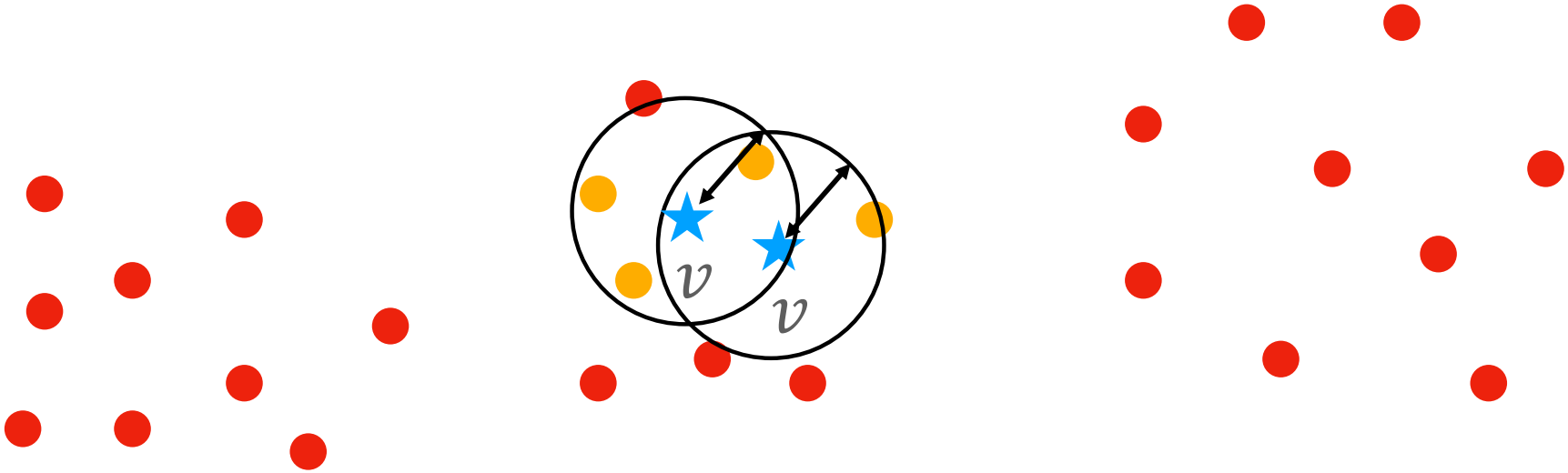


If $\mathbf{p}'$ is unvisited, 2) if the ϵ-neighbourhood of $\mathbf{p}'$ has at least $minPts$ objects, add those objects to N.

DBSCAN

$\epsilon = 5$

$minPts = 3$

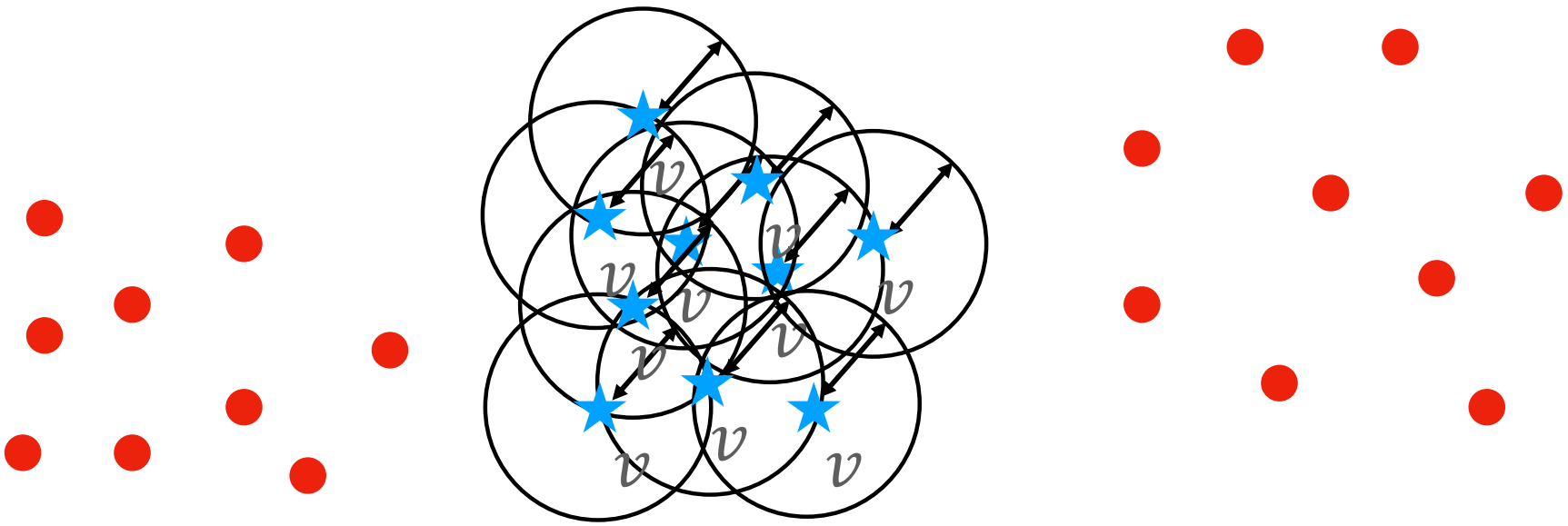


If $\mathbf{p}'$ is not yet a member of any cluster, add it to C.

DBSCAN

$\epsilon = 5$

$minPts = 3$

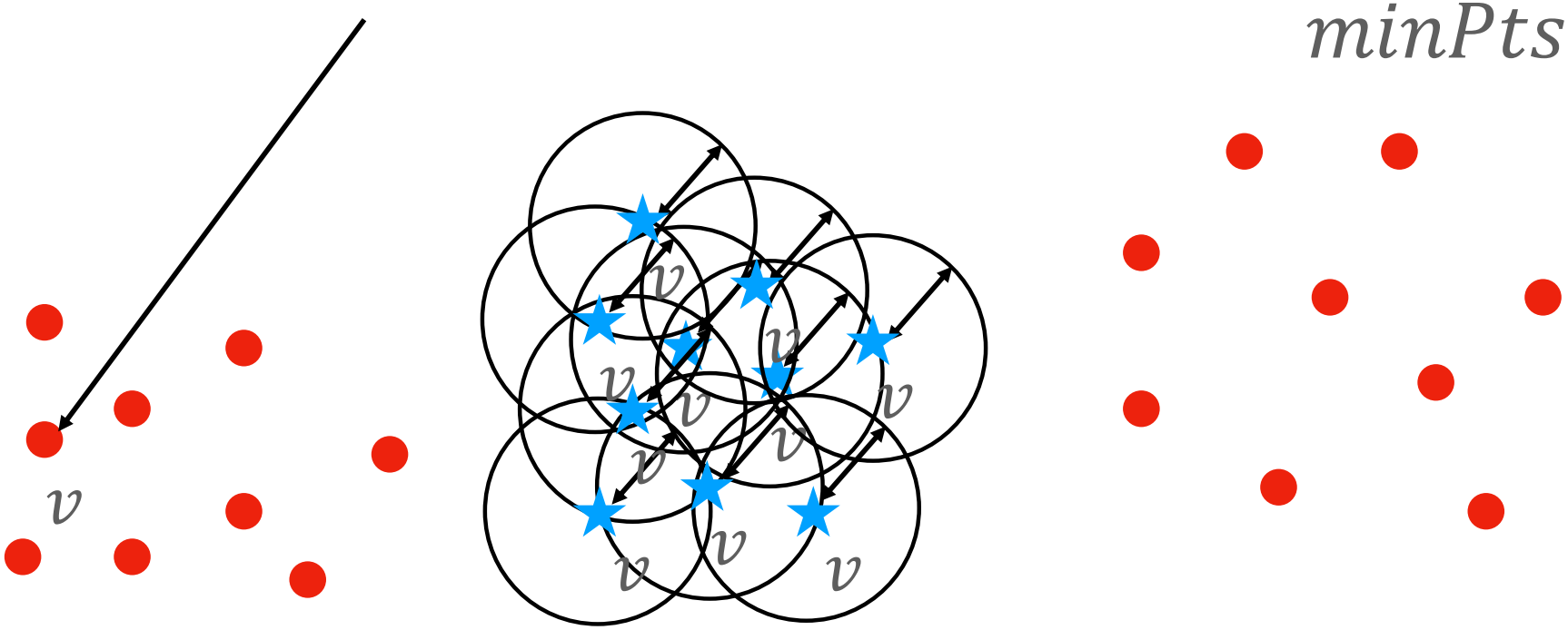


Continue for all objects in N...

DBSCAN

$\epsilon = 5$

$minPts = 3$



Then randomly choose another object, mark it visited, and repeat…

DBSCAN

Algorithm

Algorithm: DBSCAN: a density-based clustering algorithm.

Input:

- D: a data set containing n objects,
- ϵ: the radius parameter, and
- *MinPts*: the neighborhood density threshold.

Output: A set of density-based clusters.

Method:

```
mark all objects as unvisited;
do
    randomly select an unvisited object p;
    mark p as visited;
    if the ε-neighborhood of p has at least MinPts objects
        create a new cluster C, and add p to C;
        let N be the set of objects in the ε-neighborhood of p;
        for each point p′ in N
            if p′ is unvisited
                mark p′ as visited;
                if the ε-neighborhood of p′ has at least MinPts points,
                add those points to N;
            if p′ is not yet a member of any cluster, add p′ to C;
        end for
        output C;
    else mark p as noise;
until no object is unvisited;
```

Han, J., Pei, J. and Kamber, M., 2012. *Data Mining: Concepts and Techniques*. Elsevier.

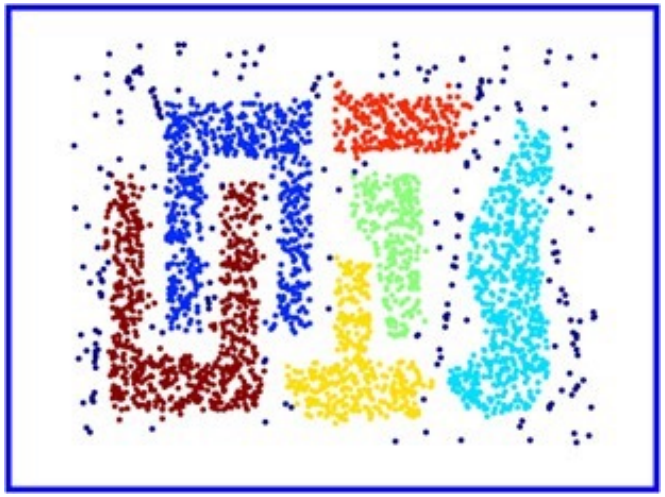
DBSCAN

- Given a data set D, a subset $C \subseteq D$ is a cluster if:
  1. For any two objects $\mathbf{o_1}, \mathbf{o_2} \in C$, $\mathbf{o_1}$ and $\mathbf{o_2}$ are density-connected, and
  2. There does not exist an object $\mathbf{o} \in C$ and another object $\mathbf{o}' \in (D - C)$ such that $\mathbf{o}$ and $\mathbf{o}'$ are density-connected.

DBSCAN

Advantages

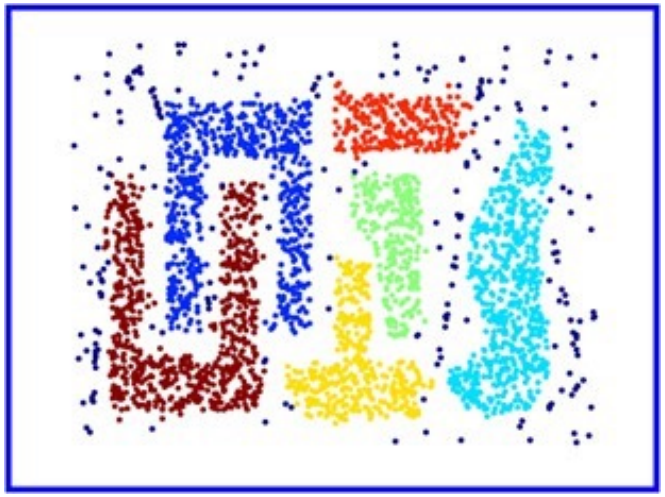
- Advantages:
  - Number of clusters is not a parameter.
  - Can find unusually shaped clusters.
  - Tolerant of noise.



DBSCAN

Disadvantages

- Disadvantages:
  - Setting parameters ϵ and $minPts$ may not be intuitive.
  - Can be slow to execute, in higher dimensions
  - Does not adapt to local variations in density.



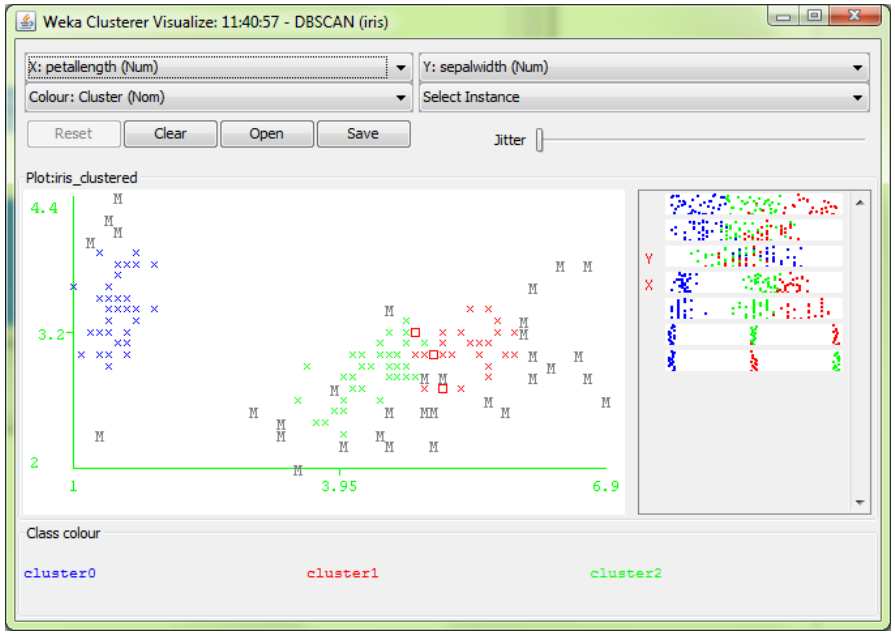
DBSCAN Evaluation in Weka

```
=== Run information ===
Scheme:weka.clusterers.DBSCAN -E 0.12 -M 5 -I
    weka.clusterers.forOPTICSAndDBScan.Databases.SequentialDatabase -D
    weka.clusterers.forOPTICSAndDBScan.DataObjects.EuclideanDataObject
Relation:     iris
Test mode:Classes to clusters evaluation on training data
DBSCAN clustering results
==========================================================================
================
Clustered DataObjects: 150
Number of attributes: 4
Epsilon: 0.12; minPoints: 5
Index: weka.clusterers.forOPTICSAndDBScan.Databases.SequentialDatabase
Distance-type:
weka.clusterers.forOPTICSAndDBScan.DataObjects.EuclideanDataObject
Number of generated clusters: 3
Elapsed time: .02
Time taken to build model (full training data) : 0.02 seconds
```

Note: experimented with many parameter choices to get ϵ = 0.12, $minPts$ = 5.

DBSCAN Evaluation in Weka

```
=== Model and evaluation on training set ===
Clustered Instances

0   45 ( 39%)

1   31 ( 27%)

2   38 ( 33%)

Unclustered instances : 36
Class attribute: class
Classes to Clusters:
  0  1  2  <-- assigned to cluster
 45  0  0 | Iris-setosa
  0  3 37 | Iris-versicolor
  0 28  1 | Iris-virginica
Cluster 0 <-- Iris-setosa
Cluster 1 <-- Iris-virginica
Cluster 2 <-- Iris-versicolor

Incorrectly clustered instances : 4.0     2.6667 %
```



Hard Clustering vs. Soft Clustering

- So far we have discussed clustering algorithms where each object can only be assigned to one cluster.
- Consider the following situation:
  - We wish to cluster University of Warwick blogs, where each cluster represents a module (CS430, CS910, CS909, CS917, CS118, …)
  - A student/staff member may write a blog about multiple modules - this blog relates to multiple clusters.
- Idea: Could a clustering algorithm allow us to assign blogs to clusters, whilst also carrying a **weight** representing the membership?

Hard Clustering vs. Soft Clustering

- **Hard Clustering**: Each data object is assigned to one (and only one!) cluster.
  - E.g.: Car is a Toyota or Honda, but cannot be both!
- **Soft Clustering**: Each data object may belong to multiple clusters, with each data object possessing a degree of membership of each cluster.
  - E.g.: Clothing colour. For example, a shirt may be red, blue, or red and blue.

Clustering Evaluation

- How do we know whether the resulting clusters are good, or just useless?
- Extreme cases: Assign all objects to a single cluster, or give each object its own cluster.
- Several methods available (depending on availability of ground-truth), but we will discuss the *Silhouette Coefficient*.

Clustering Evaluation

The Silhouette Coefficient

- Let D be a data set of n objects partitioned into k clusters, $C_1, \ldots, C_k$.
- For each object $\mathbf{o} \in D$, calculate $a(\mathbf{o})$ as the average distance between $\mathbf{o}$ and all other objects in the same cluster as $\mathbf{o}$:

$$a(\mathbf{o}) = \frac{\sum_{\mathbf{o}' \in C_i, \mathbf{o} \neq \mathbf{o}'} dist(\mathbf{o}, \mathbf{o}')}{|C_i| - 1}$$

Clustering Evaluation

The Silhouette Coefficient

- For each object $\mathbf{o} \in D$, calculate $b(\mathbf{o})$ as the minimum average distance from $\mathbf{o}$ to all clusters to which $\mathbf{o}$ does not belong:

$$b(\mathbf{o}) = \min_{C_j:1\leq j\leq k, j\neq i} \{\frac{\sum_{\mathbf{o}'\in C_j} dist(\mathbf{o},\mathbf{o}')}{|C_j|}\}$$

- The *Silhouette Coefficient* of $\mathbf{o}$ is then defined as:

$$s(\mathbf{o}) = \frac{b(\mathbf{o})-a(\mathbf{o})}{max\{a(\mathbf{o}),b(\mathbf{o})\}}$$

Clustering Evaluation

The Silhouette Coefficient

- $s(\mathbf{o})$ is between -1 and 1.
- $a(\mathbf{o})$ reflects the compactness of the cluster to which $\mathbf{o}$ belongs.
  - The smaller $a(\mathbf{o})$ is, the more compact the cluster.
- $b(\mathbf{o})$ is the degree to which $\mathbf{o}$ is separated from other clusters.
  - The larger $b(\mathbf{o})$, the more separated $\mathbf{o}$ is from other clusters.

Clustering Evaluation

The Silhouette Coefficient

- As $s(\mathbf{o})$ approaches 1, the cluster containing $\mathbf{o}$ is compact and $\mathbf{o}$ is far from other clusters (this is good!).
- If the Silhouette Coefficient is negative (i.e., $b(\mathbf{o}) < a(\mathbf{o})$), $\mathbf{o}$ is closer to objects in another cluster than the same cluster (this is bad!).
- So far, we have calculated the Silhouette Coefficient for a single object:
  - We can compute the average Silhouette Coefficient for all objects in the data set to measure the overall quality of clustering.

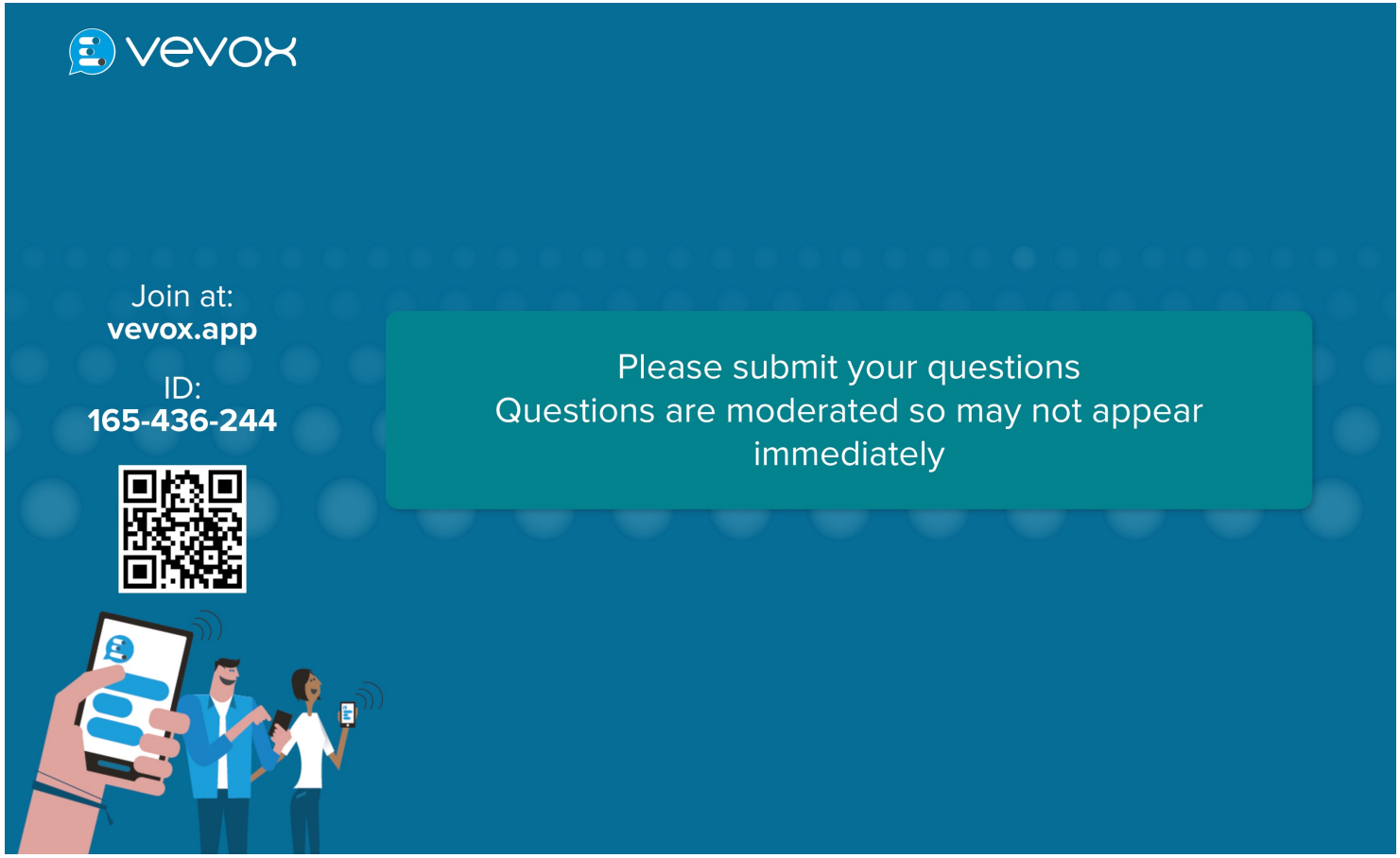
Clustering

Summary

- Clustering aims to find meaningful groups in the data.
- Many approaches to clustering, including:
  - Bottom-up: Hierarchical Agglomerative Clustering
  - Iterative: k-means
  - Density-based: DBSCAN
- We can use Weka to perform clustering, etc.
- We can analyse cluster quality using the Silhouette Coefficient.

Quiz Time

Revising our knowledge on regression, classification and clustering



Acknowledgments

- Greg Watson [Warwick, CS430/CS910]
- Han, J., Pei, J. and Kamber, M., 2012. *Data Mining: Concepts and Techniques*. Elsevier.
- Graham Cormode [Warwick, CS910]
- Florin Ciucu [Warwick, CS430/CS910]